EXPERIMENT NO. 10

Date: 30/10/2026

AIM:

To determine the I-V (voltage-current) characteristics of MOSFET (I_D vs. V_{DS}) using Keithley 4200A-SCS Parameter Analyzer.

APPARATUS:

- **Keithley 4200A-SCS Parameter Analyzer:** Core instrument with at least three Source-Measure Units (SMUs).
- **SMUs:** SMU1 (Drain), SMU2 (Source, grounded), and SMU3 (Gate).
- **Cables:** Triaxial cables (e.g., 4200-TRX-0.5 or 0.75 m) for low-noise Kelvin connections (FORCE HI, SENSE HI, GUARD).
- **Test Fixture:** Keithley 8010 or 8101-PIV test fixture for discrete devices; alternatively, Kelvin clips or banana-to-triax adapters.
- **MOSFET Device:** N-channel enhancement-mode MOSFET (e.g., 2N7000), having four terminals: Gate, Drain, Source and Body (Bulk).
- **Optional Accessories:** Ground Unit (4200-GNDU) for common reference and shielding enclosure for EMI reduction.
- **Software:** Clarius software (pre-installed on the 4200A-SCS).
- **Safety:** Compliance settings (±20 V voltage, 10 mA current); interlock for high voltages.

THEORY:

A Metal Oxide Semiconductor Field Effect Transistor (MOSFET) is a voltage-controlled semiconductor device having four terminals: Gate, Drain, Source and Body (Bulk). The drain current (I_D) is controlled by the gate-to-source voltage (V_{GS}). The body terminal is generally connected to the source terminal in a basic NMOS configuration.

The output characteristics of a MOSFET represent the relationship between drain current (I_D) and drain-to-source voltage (V_{DS}) at different fixed values of gate-to-source voltage (V_{GS}).

When V_{GS} is below the threshold voltage (V_{th}), the MOSFET remains in the cut-off region and almost no drain current flows. When V_{GS} exceeds the threshold voltage, a conducting channel is formed between the drain and source.

The MOSFET operates mainly in the following regions:

1. **Cut-off Region:** When $V_{GS} < V_{th}$, the MOSFET is OFF and the drain current is approximately zero.
2. **Linear Region:** At small values of V_{DS}, the drain current increases with V_{DS} and the MOSFET behaves like a voltage-controlled resistor.
3. **Saturation Region:** When V_{DS} reaches the saturation condition, the drain current becomes nearly constant for a fixed value of V_{GS}.

In this experiment, the Keithley 4200A-SCS Parameter Analyzer is used to apply the required voltages to the Gate and Drain terminals and measure the corresponding drain current. The source terminal is connected to ground, while the body terminal is generally connected to the source. By varying V_{DS} while keeping V_{GS} fixed at different values, the output characteristics of the MOSFET are obtained.

4. PROCEDURE

1. Connect the Keithley 4200A-SCS Parameter Analyzer to the power supply and switch on the instrument.
2. Connect the analyzer to the computer using the appropriate USB, LAN, or GPIB interface.
3. Ensure that Keithley's Clarius+ software is installed on the connected computer.
4. Connect the probes or test cables to the SMU (Source Measure Unit) outputs.
5. Connect the four terminals of the NMOS transistor (Gate, Drain, Source, and Body) to the appropriate terminals of the measurement setup, ensuring correct polarity and contact.
6. Launch the Clarius+ software and check whether it detects the Keithley 4200A-SCS analyzer.
7. Select the I-V measurement option in the software and configure the test setup.
8. Assign the SMU channels to the appropriate MOSFET terminals. Set the Source and Body terminals to ground, and apply the required Gate voltage.
9. Configure the drain voltage (V_{DS}) sweep range and measurement settings.
10. Run the test and observe the drain current (I_D) with respect to the drain-to-source voltage (V_{DS}).
11. Repeat the measurement for different gate-to-source voltage (V_{GS}) values to obtain the output characteristics of the MOSFET.
12. Save or export the measured data and plot the I_D versus V_{DS} characteristics.
13. After completing the experiment, safely stop the measurement, disconnect the DUT, and switch off the analyzer.

Note: Ensure proper grounding and suitable current-compliance settings before starting the measurement to protect the MOSFET and instrument.

5. EQUIPMENT IMAGES

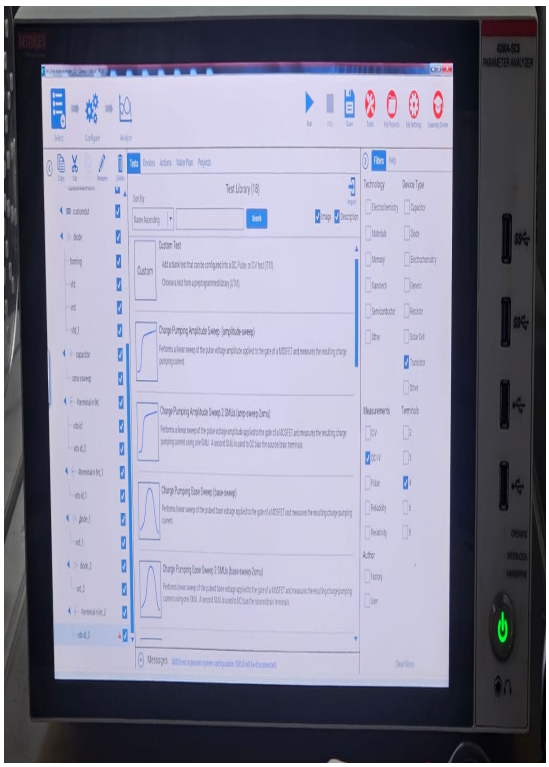


Fig. 1: Select Test Window Interface

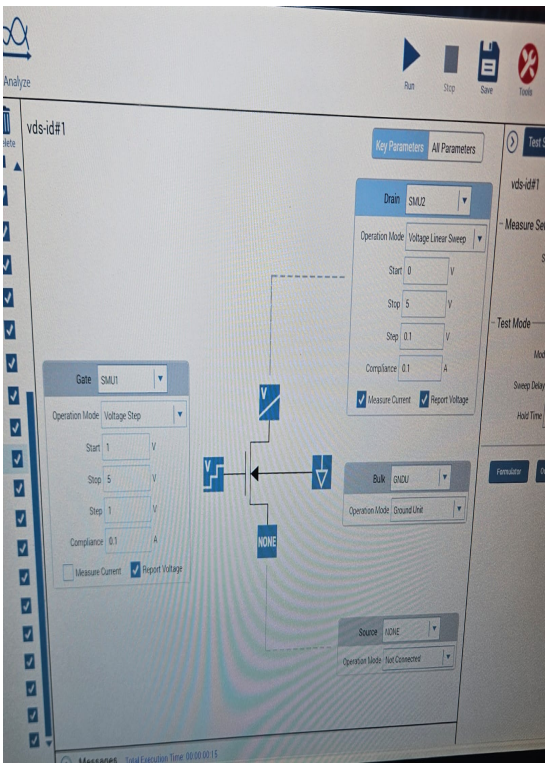


Fig. 2: Voltage Configuration Window

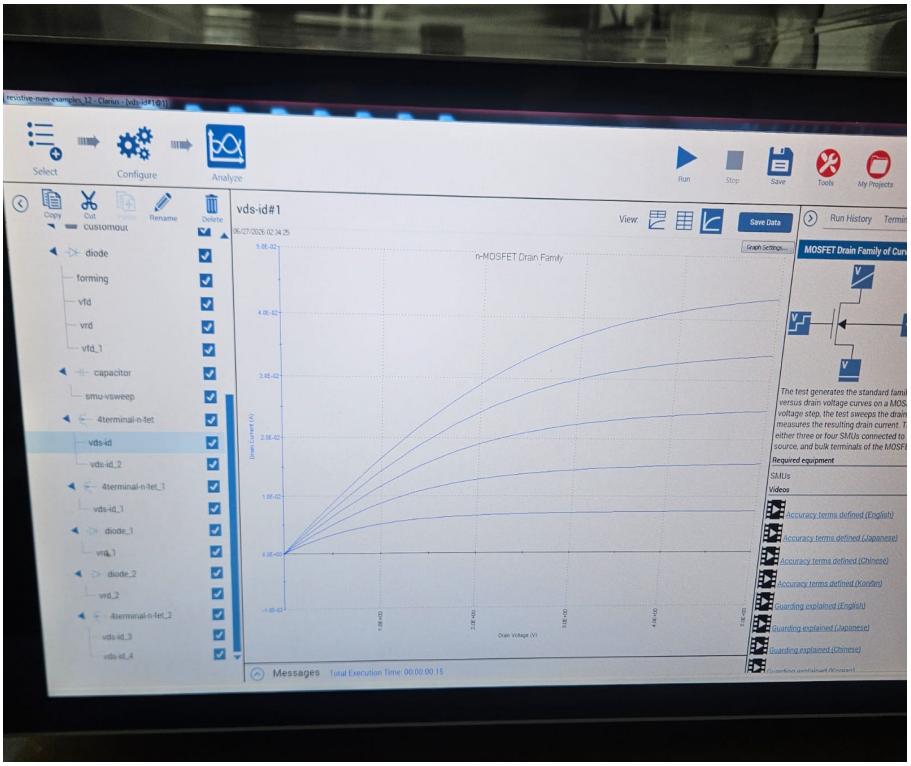


Fig. 3: Result Window Showing I_D vs. V_{DS} Characteristics

6. OBSERVATION

1. The drain current (I_D) increases with an increase in drain-to-source voltage (V_{DS}) initially.
2. After reaching a certain value of V_{DS}, the drain current becomes almost constant, indicating the saturation region.
3. The drain current increases with an increase in gate-to-source voltage (V_{GS}).
4. A family of output characteristic curves is obtained for different values of V_{GS}.

7. CONCLUSION

The voltage-current (I_D vs. V_{DS}) characteristics of an NMOS transistor were successfully studied using the Keithley 4200A-SCS Parameter Analyzer. The output characteristics were obtained for different gate-to-source voltages. The linear and saturation regions of the MOSFET were observed, and it was verified that the drain current increases with an increase in gate-to-source voltage.

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EXPERIMENT NO. 11

Date: 30/10/2026

1. AIM

To study the Raman spectroscopy characterisation of a Si sample and determine its characteristic Raman shift.

2. APPARATUS / TOOLS REQUIRED

1. Pro Raman Software
2. Pro Raman-L
3. Silicon Wafer for testing

3. THEORY

Raman spectroscopy is a non-destructive analytical technique used to study the vibrational properties and structural characteristics of materials. It is based on the interaction of monochromatic laser light with a sample. When light falls on a material, most of it is scattered without any change in energy, which is known as Rayleigh scattering. A small portion of the scattered light undergoes a change in energy, known as Raman scattering.

Raman scattering occurs due to the interaction of incident photons with the vibrational modes of the material. The difference between the incident and scattered light frequencies is called the Raman shift, which is generally expressed in inverse centimetres (cm^{-1}).

The Raman shift is calculated as:

$$\Delta\bar{\nu} = \left(\frac{1}{\lambda_0} - \frac{1}{\lambda_s}\right)$$

where,

- λ_0 = Wavelength of incident laser light
- λ_s = Wavelength of scattered light
- $\Delta\bar{\nu}$ = Raman shift in cm^{-1}

In crystalline silicon, a characteristic Raman peak is generally observed near 520 cm^{-1}. This peak corresponds to the optical phonon vibration of the silicon crystal lattice. By analysing the Raman spectrum using Pro Raman software, the characteristic Raman shift of the silicon sample can be determined.

4. PROCEDURE

1. Turn on the main power supply of the Raman spectroscopy setup.
2. Rotate the laser key to switch on the laser.
3. Adjust the laser power knob to the desired value.
4. Switch on the laptop and open the Pro Raman spectroscopy software.
5. Wait until the laser temperature drops below $60°C$, as indicated in the bottom-right corner of the software.

6. Place the silicon wafer sample in front of the laser.
7. Click on the **Single Scan** option in the top-left corner to record the Raman spectrum.
8. To remove noise and obtain the reference baseline, navigate to Configure and select **Auto Baseline**.
9. Click on the **Continuous Scan** option to scan the sample continuously.
10. Adjust the laser height using the rotary knob provided above the laser.
11. After scanning, click on Stop Scan and then select **Peak Detector** to identify the Raman peak.
12. To save the obtained data, go to File, select Save As, and save the file in either `.bitmp` or `.txt` format.
13. After completing the experiment, turn off the software.
14. Switch off the laser by rotating the key.
15. Finally, turn off the main power supply.

5. EQUIPMENT IMAGES

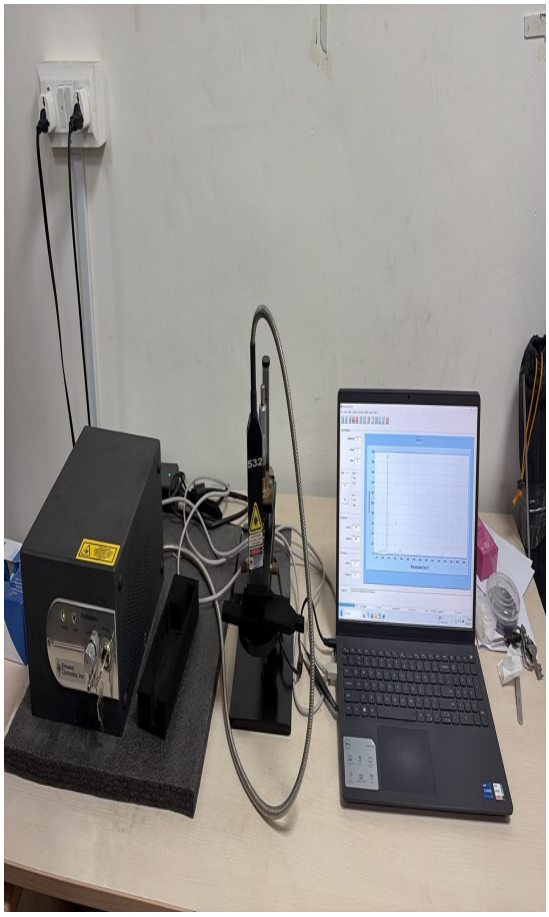


Fig. 1: Raman Spectroscopy Setup

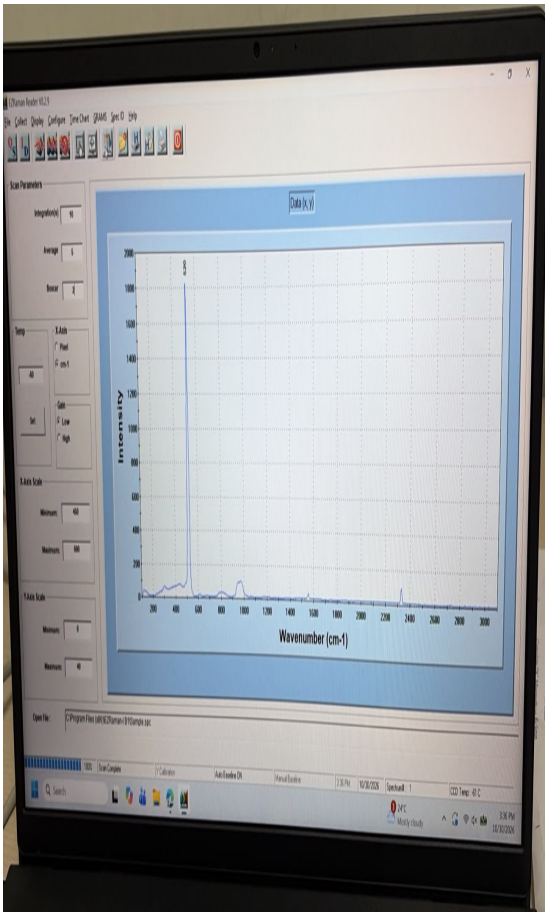


Fig. 2: Raman Spectroscopy Result

6. OBSERVATION

1. A Raman spectrum of the silicon wafer was successfully obtained using the Pro Raman software.
2. A prominent Raman peak was observed at approximately 520 cm^{-1}.
3. This peak corresponds to the characteristic optical phonon vibration of crystalline silicon.
4. The obtained spectrum shows intensity variation with respect to the Raman shift (wavenumber).

7. CONCLUSION

Raman spectroscopy characterization of the silicon wafer was successfully performed using the Pro Raman setup. The Raman spectrum showed a prominent characteristic peak near 520 cm^{-1}, which confirms the characteristic vibrational response of crystalline silicon. Thus, the characteristic Raman shift of the silicon sample was determined.

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